

SCS 3311: Mobile Application Design & Development Mini-Project

Problem Specification: Smart Home Monitoring & Control System

Project Overview

The objective of this project is to build a mobile Smart Home Monitoring and Control system consisting of a mobile application client and a companion cloud-connected hardware simulation system. The system allows users to interact with multiple floor plans, monitor and toggle equipment with differing specialized capabilities (e.g., individual power outlets, multi-switch units, lighting, safety-critical devices like irons, and security cameras), and execute automated backend-driven safety rules.

Core Functional Requirements

Multi-Floor Interactive Dashboard (Mobile Client)

- **Floor Plan Layouts:** The app must support adding and managing multiple house-floor plans. Users should be able to view an abstract(simple) grid mapping overlaid onto specific floor layouts. You can use some free sample plans for demo purpose.
- **Device Control UI:** Toggling device states must reactively update on the user interface. Devices must display their current operational status (`ON` , `OFF` , `ERROR` , `DISCONNECTED`).

Heterogeneous Device Profiles

- **Electrical Outlets:** Simple single-node binaries representing a continuous power supply.
- **Multi-Switch Units:** A single physical gang-box unit managing a variable number of separate, individually addressable switches (e.g., a 2,3 or 5 switch unit mapped to a single entity in the system).
- **Scheduling of Operation:** Specialized slots assigned to appliances prone to fire hazards (e.g., clothing irons). These must support a maximum permissible active duration configuration. Also some light bulbs may be set to turn on and off automatically during a preset time period.
- **Security Cameras:** Interface elements dedicated to monitoring spaces via mock camera snapshots / mock uri streams.

Online Synchronization

- **Bidirectional Sync:** State changes made in the mobile application must reflect quickly in the cloud database, and any state updates driven externally must quickly update the mobile viewport without manual refresh triggers.

- **Server-Side Safety Cutoffs:** To protect life and property, safety-critical slots may be regulated by a backend cloud listener or a worker process. If a high-power device configuration's *max_on_duration* is breached, an automatic flip of the database state to `OFF` and push of an alert is important.
- **Reporting:** Usage data of the important devices can be tracked from the mobile app. You may decide the best way to present this information to the user.

Companion Hardware Simulator

- To bypass physical component constraints while maintaining a realistic IoT development experience, you may deploy a web-based **Hardware Simulator Dashboard**. This simulator represents the physical home appliances, listening directly to database updates and visually reflecting changes online.

Assessment & Submission Deliverables

Evaluation will be based on both group cohesion and individual defense.

1. **Source Code & Git:** Hosted on a platform like GitHub and link to final apk must be shared.
2. **Technical Documentation:** A concise report outlining the synchronizing mechanism, floor representation and simulator operations.
3. **Recorded Video for Demonstration:** All three members must present in the video with presenter **introduction** and their **contribution**. Do not exceed 25 minutes of video.